Eugenio Dellepiane

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Summary

I am a mathematician currently in my second postdoctoral position. My research interests lie in mathematical analysis, especially in complex analysis and operator theory. I am passionate about teaching and I am actively involved in the international mathematical community: I regularly attend conferences, workshops, and seminars, where I value the opportunity to exchange ideas and and build new collaborations. This CV was updated on November 8th, 2026.

Education & Research

Sep 2026 – ongoing
Postdoc at University of Twente (NL)
Under the supervision of Felix Schwenninger.

Nov 2025 – Aug 2026
Postdoc at Université Laval, in Québec City (CA)
Under the supervision of Javad Mashreghi. This position was supposed to last until December 2026. It ended sooner because of teaching duties in the new position at Twente.

Mar – Aug 2025
Postdoctoral scholarship at Politecnico di Torino (IT)
I won a five months scholarship to work on the research project titled *"Hardy subspaces, their structure and computational analysis"*, under the supervision of Anita Tabacco.

Oct 2021 – Jan 2025
Ph.D. in Mathematics at Università di Milano - Statale (IT).
Thesis title: *An analysis of de Branges–Rovnyak and Dirichlet spaces.*

Sep 2019 – Oct 2021
M.Sc. in Pure Mathematics at Università di Genova (IT).
Grade: 110/110 cum laude.
Thesis title: *Singular integrals and functional calculus for divergence-type operators.*

Sep 2016 – Jul 2019
B.Sc. in Mathematics at Università di Genova (IT).
Grade: 110/110 cum laude.
Thesis title: *Haar measures on topological groups and the continuous wavelet transform.*

Research publications

Journal articles

2026

C. Bellavita, E. Dellepiane, and G. Stylogiannis, "Rhaly operators on weighted Hardy spaces and factorable matrices," *J. Geom. Anal.*, vol. 36, Article no. 413, 2026. DOI: https://doi.org/10.1007/s12220-026-02666-9

E. Dellepiane and D. Seco, "Orthogonal polynomials in de Branges-Rovnyak spaces," *J. Math. Anal. Appl.*, vol. 565, no. 1, Paper No. 130974, 20, 2027. DOI: https://doi.org/10.1016/j.jmaa.2026.130974

C. Bellavita, E. Dellepiane, A. Hartmann, and J. Mashreghi, "Infinitely supported harmonically weighted Dirichlet spaces which are de Branges–Rovnyak spaces," *J. Funct. Anal.*, vol. 291, no. 8, Paper No. 111558, 47 pp. 2026. DOI: https://doi.org/10.1016/j.jfa.2026.111558

E. Dellepiane, J. Mashreghi, M. Nasri, and W. Verreault, "The Cesàro operator on local Dirichlet spaces," *Indag. Math. (N.S.)*, vol. 37, no. 5, pp. 1883–1896, 2026. DOI: https://doi.org/10.1016/j.indag.2026.02.005

2025

- C. Bellavita, E. Dellepiane, and G. Stylogiannis, "Boundedness, compactness and Schatten class for Rhaly matrices," *J. Lond. Math. Soc.*, vol. 112, no. 4, Paper no. e70304, 36 pp, 2025. DOI: https://doi.org/10.1112/jlms.70304
- E. Dellepiane, M. M. Peloso, and A. Tabacco, "On the equality of de Branges–Rovnyak and Dirichlet spaces," *Complex Anal. Oper. Theory*, vol. 19, no. 119, 2025. DOI: https://doi.org/10.1007/s11785-025-01747-x
- C. Bellavita, E. Dellepiane, and J. Mashreghi, "The spectral analysis of the difference quotient operator on model spaces," *Bull. Sci. Math.*, vol. 204, no. Paper no. 103673, 22 pp. 2025. DOI: https://doi.org/10.1016/j.bulsci.2025.103673
- C. Bellavita and E. Dellepiane, "Embedding model and de Branges–Rovnyak spaces in Dirichlet spaces," *Complex Var. Elliptic Equ.*, vol. 70, no. 2, pp. 228–246, 2025. DOI: https://doi.org/10.1080/17476933.2023.2298837

Preprints

- F. J. Cruz-Zamorano, E. Dellepiane, and D. Seco, *Compact and Hilbert–Schmidt composition operators on weighted Hardy spaces*, 2026. arXiv: 2610.40148 [math.FA]. URL: https://arxiv.org/abs/2610.40148
- C. Bellavita, E. Dellepiane, and S. Luo, *A characterization of equality between de Branges–Rovnyak and Dirichlet spaces*, 2026. arXiv: 2610.36943 [math.FA]. URL: https://arxiv.org/abs/2610.36943
- E. Dellepiane, A. Monguzzi, and M. Monti, *Hardy spaces of discrete holomorphic functions on the upper half-lattice*, 2026. arXiv: 2607.17726 [math.CV]. URL: https://arxiv.org/abs/2607.17726

Conferences & Seminars

Invited speaker

- **VI Young Researchers Workshop in Harmonic Analysis**
  Università di Milano - Bicocca, 11 - 13 Jan 2027
  I am invited to give a 30 minutes talk.
- **International Workshop on Operator Theory and its Applications (IWOTA26)**
  Université Laval, 3 - 7 August 2026
  I was invited to give two 25 minutes talk, during the special sessions "Operator theory and function spaces" and "Invariant subspaces and cyclicity".
- **CRM-Montreal-Québec Analysis Seminar**
  Université Laval, 14 Nov 2025
  I was invited to give a 1h seminar titled *"Boundedness, compactness and Schatten class for Rhaly matrices"*.
- **Seminario de Análisis Matemático y Matemática Aplicada**
  Universidad de La Laguna, 16 Oct 2025
  I was invited to give a 1h seminar titled *"The difference quotient operator on model spaces"*. I gave the talk in spanish.
- **Séminaire d'Analyse**
  Université de Bordeaux, 2 Oct 2025
  I was invited to give a 1h seminar titled *"Infinitely supported harmonically weighted Dirichlet spaces which are de Branges–Rovnyak spaces"*.
- **Workshop in Complex and Harmonic Analysis**
  Università di Milano - Statale, 16 - 17 September 2025
  I was invited to give a 50 minutes talk, titled *"Difference quotient operators and embeddings between spaces of analytic functions"*.

Conferences & Seminars (continued)

- **International Workshop on Operator Theory and its Applications (IWOTA25)**
  University of Twente, 14 - 18 July 2025
  I was invited to give a 25 minutes talk, during the special session on operator theory on function spaces in memory of N. Vasilevski, titled *"Boundedness, compactness and Schatten class for Rhaly matrices"*.
- **5th Young Researchers in Functional Analysis and Mathematical Physics meeting**
  University of Aveiro, 8 - 11 July 2025
  I was invited to give a 20 minutes talk, titled *"Boundedness, compactness and Schatten class for Rhaly matrices"*.
- **Analysis on Complex and Discrete Structures**
  Università di Milano - Statale, 20 - 21 February 2025
  I was invited to give a 1h talk titled *"Orthonormal bases in model spaces"*.
- **IV Young Researchers Workshop in Harmonic Analysis**
  Università di Milano - Statale, 22 - 24 Jan 2025
  I was invited to give a 30 minutes talk titled *"Hadamard multipliers and polynomial approximation in Dirichlet spaces"*.
- **Séminaire "Analyse fonctionnelle"**
  Université de Lille, 4 Oct 2024
  I was invited to give a 1h talk titled *"Difference quotient operators and embeddings between spaces of analytic functions"*.
- **III Young Researchers Workshop in Harmonic Analysis**
  Università di Genova, 10 – 12 Jan 2024
  I was invited to give a 30 minutes talk titled *"Embedding model and de Branges-Rovnyak spaces in Dirichlet spaces"*.

Contributed talks

- **Canadian Mathematical Society Winter Meeting 2025**
  Toronto, 5 - 8 Dec 2025
  I gave a contributed talk titled *"Boundedness, compactness and Schatten class for Rhaly matrices"*.
- **Advanced Courses in Operator Theory and Complex Analysis (ACOTCA25)**
  Université Clermont Auvergne, Clermont Ferrand, 16 - 20 Jun 2025
  I gave a contributed talk titled *"$H(b)$ spaces that are $\mathcal{D}_\mu$ spaces, with infinitely many atoms"*.
- **XLIV Convegno Nazionale di Analisi Armonica**
  Università di Napoli Parthenope, 27 - 30 May 2025
  I gave a contributed talk titled *"Boundedness, compactness and Schatten class for Rhaly matrices"*.
- **Advanced Courses in Operator Theory and Complex Analysis (ACOTCA24)**
  Universidad de La Laguna, Tenerife, 24 – 28 Jun 2024
  I gave a contributed talk titled *"On the difference quotient operator on model spaces"*.
- **Spaces of Analytic Functions and their Operators**
  Saarland University, Saarbrücken, 13 – 17 May 2024
  I gave a contributed talk titled *"Embedding model and de Branges-Rovnyak spaces in Dirichlet spaces"*.
- **Symposium in Harmonic and Complex Analysis, Microlocal and Geometrical Analysis**
  Università di Bologna, 24 – 26 Jan 2024
  I gave a contributed talk titled *"Embedding Model Spaces in Dirichlet spaces"*.

Others

- **Complex Analysis and Spaces of Holomorphic Functions**
  Würzburg, 21 - 25 Sep 2026

Conferences & Seminars (continued)

- **Complex Analysis and Operator Theory: a Series of 3-Hour Lecture**
  Milano, 7 - 11 Sep 2026
- **Workshop on Operator Theory, Complex Analysis and Applications (WOTCA25)**
  University of Aveiro, 7 - 11 July 2025
- **Hilbert Function Spaces 2025**
  Villa Mondragone (Frascati), 30 Jun - 4 July 2025
- **Operators on analytic function spaces**
  CIRM - Luminy (Marseille), 2 – 6 Dec 2024
- **Advanced Courses in Operator Theory and Complex Analysis (ACOTCA23)**
  Aristotle University, Thessaloniki, 26 – 30 Jun 2023
- **XLII Convegno Nazionale di Analisi Armonica**
  Università di Milano - Bicocca, 29 – 31 May 2023
- **Young Researchers Workshop in Harmonic Analysis**
  Politecnico di Torino, 19 – 20 Jan 2023
- **(New trends in) complex and Fourier analysis, and operator theory**
  Workshop INdAM in Rome, Italy, 12 – 16 Sep 2022
- **11th International Conference on Harmonic Analysis and Partial Differential Equations**
  ICMAT conference at El Escorial, Madrid, 6 – 10 Jun 2022
- **Geometric Aspects of Complex and Harmonic Analysis**
  Università di Bologna, 17 – 18 Jan 2022
- **8th European Congress of Mathematics**
  Portorož, Slovenia, 19 – 25 Jun 2021

Teaching

Feb 2027 – May 2027 — **Lecturer**
University of Twente
I am responsible for lectures and tutorials for the course "Mathematics 3" in the degree of Creative Technology.

Nov 2026 – Feb 2027 — **Lecturer**
University of Twente
I am responsible for lectures and tutorials for the course "Mathematics 2" in the degree of Creative Technology.

Sep 2026 – Feb 2027 — **Teaching assistant**
University of Twente
I am running the tutorials for the course "Applied Functional Analysis" in the Master degree in Mathematics.

Sep 2024 – Feb 2025 — **University Tutor**
Università di Milano - Statale
Assisted with exercises for the course "Functional Analysis" in the Master degree in Mathematics.

Teaching (continued)

Sep 2023 – Feb 2024
University Tutor
Università di Milano - Statale
Assisted with exercises for the course "Analisi Matematica 3" in the undergraduate degree in Mathematics.

University Tutor
Università di Milano - Statale
Assisted with exercises for the course "Analisi Matematica 1" in the undergraduate degree in Physics.

Feb 2024 – Jun 2024
University Tutor
Università di Milano - Statale
Assisted with exercises for the course "Analisi Matematica 2" in the undergraduate degree in Physics.

Sep 2022 – Feb 2023
Teaching assistant
Politecnico di Milano
Prepared and taught exercise classes for "Analisi Matematica 1" in the undergradute degree in Mechanical Engineering.

Feb 2022 – Jun 2022
University Tutor
Politecnico di Milano
Assisted with exercises for the course "Analisi Matematica 2" in the undergraduate degree in Electrical Engineering.

Mar 2021 – Jul 2021
University Tutor
Università di Genova
Assisted students with exercises and mock exams for the course "Matematica generale" in the undergraduate degree in Natural Sciences.

Miscellaneous

Periods abroad

Apr – Jun 2024 — **Visiting period (Ph.D.)**. I spent three months in Université Laval in Québec City (CA), working with Javad Mashreghi.

Sep 2020 – Feb 2021 — **Erasmus (Master)**. I spent a semester in Universidad Complutense de Madrid, where I passed 4/4 exams in Spanish language.

Jan – Jun 2019 — **Erasmus (Bachelor)**. I spent a semester in Universidad Complutense de Madrid, where I passed 4/4 exams in Spanish language.

Languages

- Italian C2 (native language)
- Spanish C1
- English C1
- French B2

Miscellaneous (continued)

Ph.D. courses and summer schools

Mar 2025 — **Bounded and unbounded operators on Hilbert spaces**
25 hours Ph.D. course held at Politecnico di Torino by Alessio Martini on spectral theory for bounded and unbounded operators on Hilbert spaces, including the theory of commutative C^*-algebras.

Nov 2024 — **Modern singular integral theory: sparse domination**
12 hours Ph.D. course held at Politecnico di Torino by José Conde Alonso (Universidad Autónoma, Madrid).

May 2023 — **Function spaces on the disk and the plane: Model and de Branges spaces, recent results, extensions, and applications**
Two weeks of advanced courses in Cortona, Tuscany, at the Palazzone of the Scuola Normale Superiore. Classes held by William T. Ross and Marco M. Peloso.